

Retina

→ Sheet of neurons → 5 major cell types → 3 cellular layers.

→ Photoreceptor cells → phototransduction

↓ to

bipolar cells (innermost layer)

retinal layer → low-level visual processing

• Blur is smallest near the optical axis

• Fovea → Area of retina near the optical axis → Vision is sharpest
center of our gaze

• Density of photoreceptor, bipolar cells and ganglion cells is highest

In fovea (center of fovea) the other cellular layers are pushed aside

• Optic Disc → where axons of retinal ganglion cells converge → optic nerve

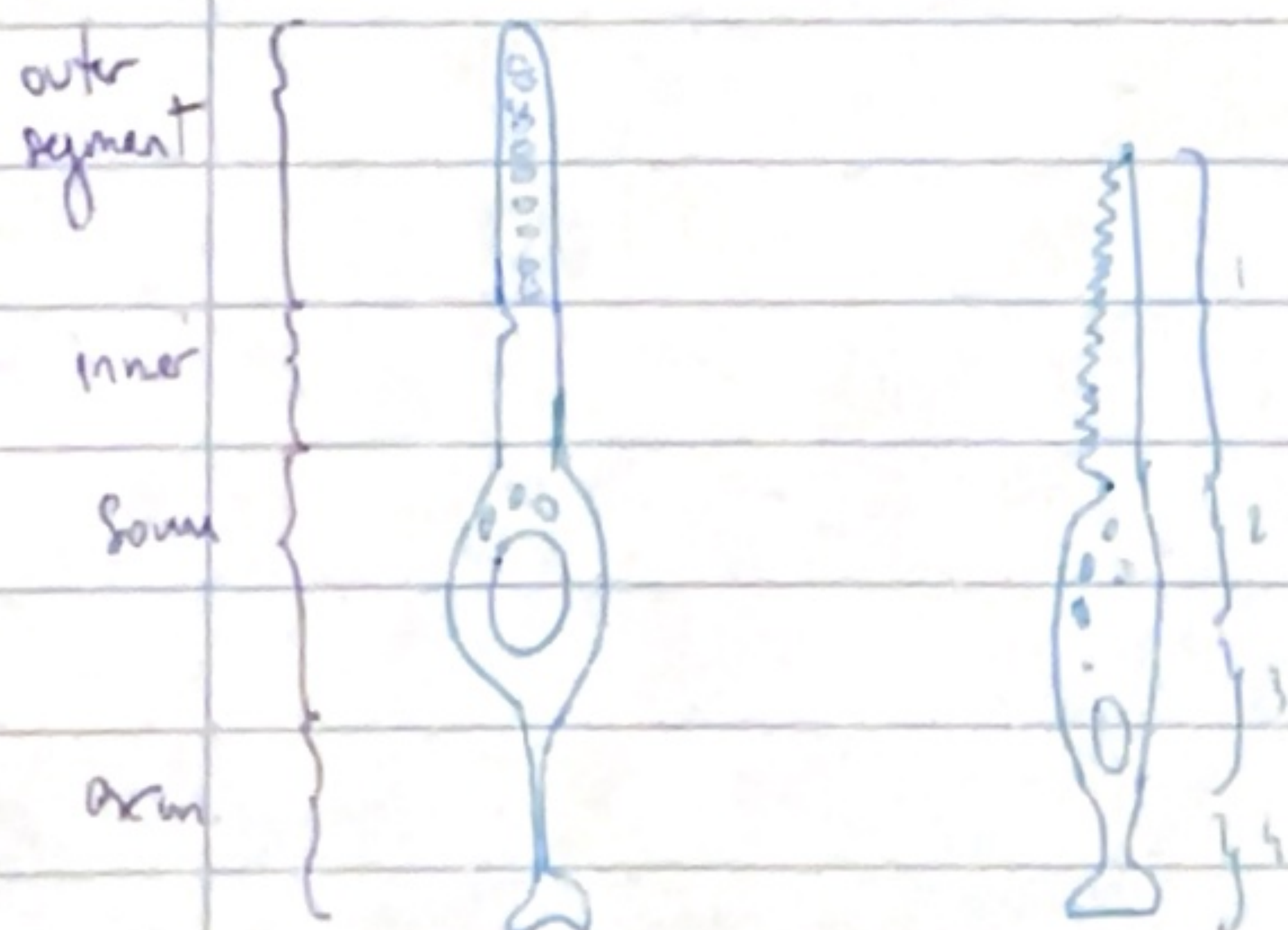
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devoid of photoreceptors → blind spot

RODS AND CONES

Photoreceptor cells

- outer segment
- inner
- cell body
- synaptic terminal



histologically different ≠

Bastones (Rods) → Vision under dim illumination
≠ intensidades de luz

CONES

- less sensitivity to light
- responsible for vision in daylight
- faster than rods

S (short wave) M (medium wave) L (long wave)

Human retina $\left\{ \begin{array}{l} 100 \text{ million rods} \\ 6 \text{ million cones} \end{array} \right\} \neq \text{distributed}$

$\Rightarrow$ Central fovea — No rods. $\Rightarrow$ Retinal Center = Daylight Vision

$\Rightarrow$ All photoreceptors ^{become} larger and $\uparrow$ spaced toward the periphery of the retina